



Tutorial exercises (3)

1. EXERCISE

Write an algorithm that allows you to:

1. Ask for two positive integers, then calculate the first to the power of the second.
2. Show the divisors of an integer N.
3. Check if a number is prime or not.

2. EXERCISE

Write an algorithm that asks the user to provide a positive integer less than 100 until the answer is satisfactory. The dialogue with the user will look like this :

Give a positive integer less than 100 :

453

Please, less than 100 :

-5

Please, positive integer:

15

thanks for the number **15**

- **Using repeat loop , then while loop**

3. EXERCISE

A perfect number is a number that is equal to the sum of all its divisors except itself.

- Build an algorithm to search for all perfect numbers between 1 and 1000.

4. EXERCISE

Show the execution history of the **mystery** algorithm and deduce the final value of **S**.

```
Algorithm Mystere ;
Var i, j, s : Integer ;
Begin
    s ← 0 ;
    For i=1 TO 4 Do
        j ← 1 ;
        While j ≤ 5 Do
            s ← s+1 ;
            j ← j+1 ;
        Endwhile ;
    Endfor ;
    Write (s);
End.
```

5. EXERCISE

Let **X** be a binary number consisting of **N** digits.

- Write the algorithm that displays the decimal value corresponding to this binary number.

6. EXERCISE

Let **X** be a positive integer given by the user.

- Write an algorithm that displays **in order all the digits that make up this number**, and also displays the number composed of the digits of **X** in **reverse order**.

7. EXERCISE

In an oral examination, the candidates are examined one after the other by an examiner, who asks each candidate questions and then gives them a mark.

After each candidate has passed, the examiner takes **3 minutes** to give the mark and start with the next candidate.

- If the exam starts at time **T (H:M:S)**, write the algorithm that displays the start and end times for each candidate (the number of candidates and the duration of each candidate's exam are given by the user).

8. EXERCISE

A medical laboratory carries out analyses for patients. The result of each analysis is obtained after a certain number of days, depending on the nature of the analysis.

1. Write an algorithm that displays the date of the result of an analysis based on the date of the analysis. (The number of days required for the result of this analysis is given by the user).
2. Repeat the previous question using the algorithm that calculates the next day's date.